SOURCE: MSC.141(76).pdf
requirements should be satisfied:

.1 heading control lines, intended for minor adjustment, should be located at the centre line of the stem and stern, in a sy mmetrical fashion and at a level between the position of KG and the damaged waterline; and

.2 the carriage speed should be equal to the actual drift speed of the model with speed adjustment made when necessary.

4.3 At least 10 experiments should be carried out . The test period for each experiment should be of a duration such that a stationary state is reached, but not less than 30 min in full-scale. A different wave realisation train should be used for each experiment.

5 Survival criteria

5.1 The model should be considered as survivi ng if a stationary state is reached for the successive test runs required in 4.3.

5.2 The model should be considered as capsized if angles of roll of more than 30° to the vertical axis or steady (average) heel greater than 20° for a period longer than 3 minutes full-scale occur, even if a stationary state is reached.

6 Test documentation

6.1 The model test programme should be approved by the Administration in advance.

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SOURCE: MSC.141(76).pdf
full-scale occur, even if a stationary state is reached.

6 Test documentation

6.1 The model test programme should be approved by the Administration in advance.

6.2 Tests should be documented by means of a report and a video or other visual records containing all relevant information on the model and the test results, which are to be approved by the Administration. These should include, as a minimum, the theoretical and measured wave spectra and statistics (HS, TP, TZ) of the wave elevation at the 3 different locations in the basin for a representative realisation, and for the tests with the model, the time series of main statistics of the measured wave elevation close to the wave maker and records of model roll, heave and pitch motions, and of the drift speed.

GUIDANCE NOTES ON THE REVISED MODEL TEST METHOD

The purpose of these notes is to ensure uniform ity in the methods employed in the construction

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ATTACHMENT

GUIDANCE NOTES ON THE REVISED MODEL TEST METHOD

The purpose of these notes is to ensure uniform ity in the methods employed in the construction and verification of the model as well as the undertaking and analyses of the model tests, while appreciating that available facilities and costs will affect in some way this uniformity.

The contents of paragraphs 1 and 2 of th e Revised Model Test Method are considered self-explanatory.

Paragraph 3 - Ship model

3.1 The material of which the model is made is not important in itself, provided that the model, both in the intact and damaged condition, is sufficiently rigid to ensure that its hydrostatic properties are the same as those of the actual ship and also that the flexural response of the hull in waves is negligible.

It is also important to ensure that the damaged compartments are modelled as accurately as practicably possible to ensure that the correct volume of floodwater is represented.

Since ingress of water (even small amounts) in to the intact parts of the model will affect

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SOURCE: MSC.141(76).pdf
as practicably possible to ensure that the correct volume of floodwater is represented.

Since ingress of water (even small amounts) in to the intact parts of the model will affect its behaviour, measures must be taken so that this ingress does not occur.

In model tests involving worst SOLAS damage s near the ship ends, it was observed that progressive flooding was not possible because of the tendency of the water on deck to accumulate near the damage opening and hence fl ow out. As such models were able to survive very high sea states, while they capsi zed in lesser sea states with less onerous SOLAS damages, away from the ends, the limit 35% was introduced to prevent this.

Extensive research carried out for the purpos e of developing appropriate criteria for new vessels has clearly shown that in addition to the GM and freeboard being important parameters in the survivability of passenger ships, the area under the residual stability

curve is also another major factor. C onsequently in choosing the worst SOLAS damage for compliance with the requirement of paragr aph 3.5.1 the worst damage is to be taken

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curve is also another major factor. C onsequently in choosing the worst SOLAS damage for compliance with the requirement of paragr aph 3.5.1 the worst damage is to be taken as that which gives the least area under the residual stability curve.

3.2 Model particulars

.1 In recognising that scale effects play an important role in the behaviour of the model during tests it is important to ensure that these effects are minimised as much as practically possible. The mode l should be as large as possible since details of damaged compartments are easie r constructed in larger models and the scale effects are reduced. It is therefore required that the model length is not less than that corresponding to 1:40 scale or 3 m, whichever is greater.

It has been found during tests that the vertical extent of the model can affect the results when tested dynamically. It is th erefore required that the ship be modelled to at least three superstructure standard heights above the bulkhead (freeboard) deck so that the large waves of the wave train do not break over the model.

RESOLUTION MSC.141(76)
(adopted on 5 December 2002)
REVISED MODEL TEST METHOD UNDER RESOLUTION 14

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SOURCE: MSC.141(76).pdf
deck so that the large waves of the wave train do not break over the model.

RESOLUTION MSC.141(76)
(adopted on 5 December 2002)
REVISED MODEL TEST METHOD UNDER RESOLUTION 14
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.2 The model in way of the assumed dama ges should be as thin as practically possible to ensure that the amount of floodwater and its centre of gravity are adequately represented. The hull thickness should not exceed 4 mm. It is recognised that it may not be possible for the model hull and the elements of primary and secondary subdivision in way of the damage to be constructed with sufficient detail and due to these constr uctional limitations it may not be possible to calculate accurately the assumed permeability of the space.

.3 It is important that not only the draught s in the intact condition are verified but also that the draughts of the damaged model are accurately measured for correlation with those derived from th e damaged stability calculation. For practical reasons a tolerance of +2 mm in any draught is accepted.

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SOURCE: MSC.141(76).pdf
also that the draughts of the damaged model are accurately measured for correlation with those derived from th e damaged stability calculation. For practical reasons a tolerance of +2 mm in any draught is accepted.

.4 After measuring the damaged draughts it may be found necessary to make adjustments to the permeability of the damaged compartment by either introducing intact volumes or by adding weights. However, it is also important to ensure that the centre of gravity of the floodwater is accurately represented. In this case any adjustments made must err on the side of safety.

If the model is required to be fitted with barriers on deck and their height is less than the bulkhead height indicated below, the model should be fitted with CCTV so that any "splashing over" and a ny accumulation of water on the undamaged area of the deck can be monitored. In this case a video recording of the event should form part of the tests records.

The height of transverse or longitudinal bulkheads which are taken into account as effective to confine the assumed accumu lated sea water in the compartment

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